Public Awareness and Expectations of Accelerated & Reliable MRI Imaging Using Machine Learning

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Abstract

Magnetic Resonance Imaging (MRI) is widely used in diagnosing various conditions, there are many disadvantages associated with this technology, such as long imaging time, patient discomfort, and expensive costs. This research paper will explore the level of general awareness of the public regarding the ability of machine learning technologies to improve the functionality and accuracy of the diagnostics tool. To achieve this goal, a survey was conducted via Google Forms among 60 respondents of different age groups and educational backgrounds. The results revealed that most respondents have already experienced MRI and had an understanding of the possibility of the application of machine learning technologies. Respondents thought that ML would help to reduce time and improve image quality, and 91.1% were ready to use ML-supported MRI. However, concerns for data privacy were present.

Methodology

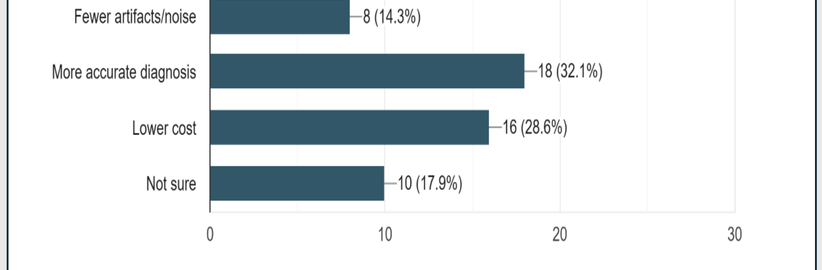
This study used a descriptive cross-sectional survey to assess public awareness and trust regarding the use of Machine Learning (ML) in MRI procedures. A total of 56 participants from different age groups and educational backgrounds were selected. Data was collected through a structured questionnaire distributed via Google Forms.

The survey covered five main areas: personal MRI experience, ML literacy, expectations from ML-based MRI (such as speed, image quality, and accuracy), public trust in ML-assisted diagnosis, and concerns about data privacy and security. Most participants were aged between 25–40 years and represented a relatively young adult population.

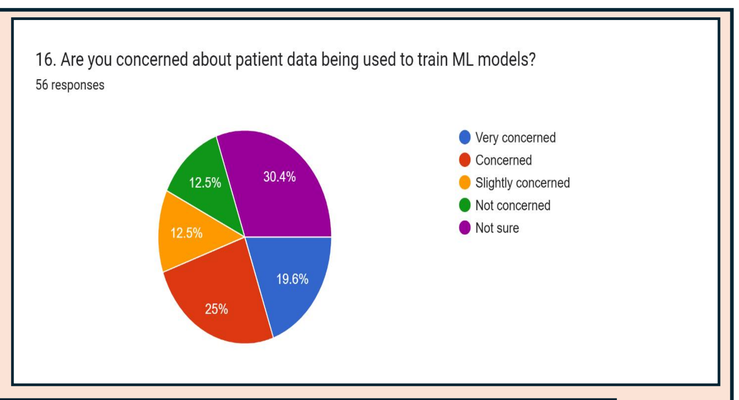
Introduction

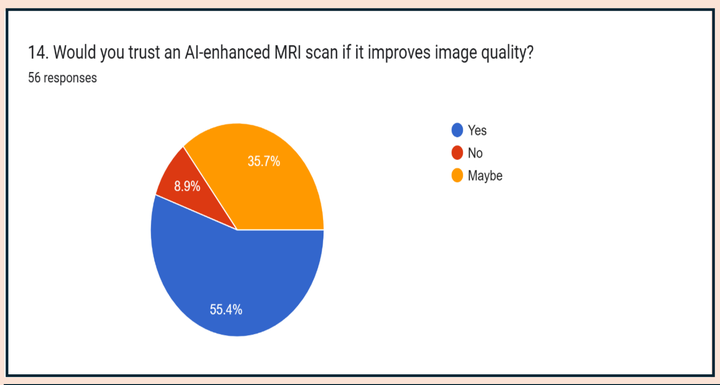
The Magnetic Resonance Imaging (MRI) is one of the diagnostic imaging methods which plays a critical role in the process; yet, due to its complicated scanning process, the high cost of the method, and patients' pain while undergoing the procedure, there are several problems associated with it.

The development of innovative techniques like machine learning can be used to solve this problem by making the scanning procedure faster and ensuring the accuracy of decision-making through improved image resolution.



Expected improvements from an ML-enhanced MRI system Faster scans was the most expected improvement (46.4%), followed by more accurate diagnosis (32.1%) and lower cost (28.6%). This indicates that respondents prioritize speed and diagnostic accuracy as the key benefits of ML integration in MRI.

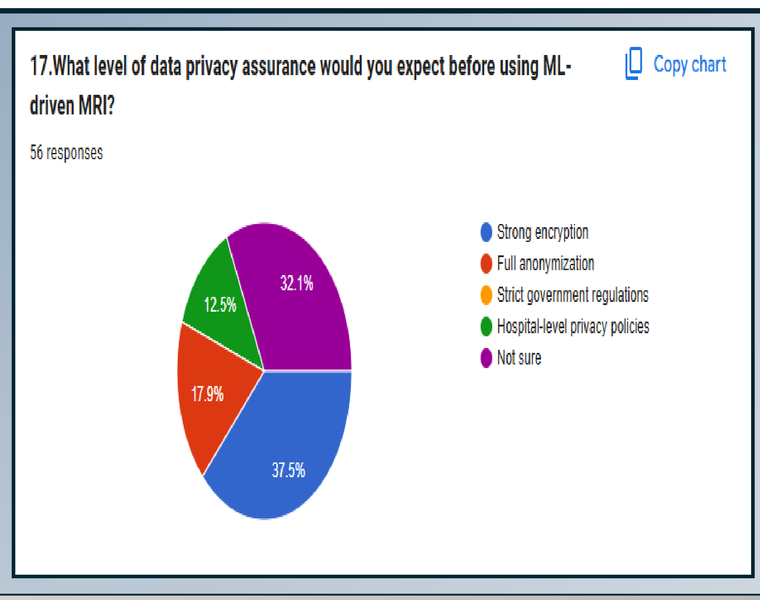




Result:

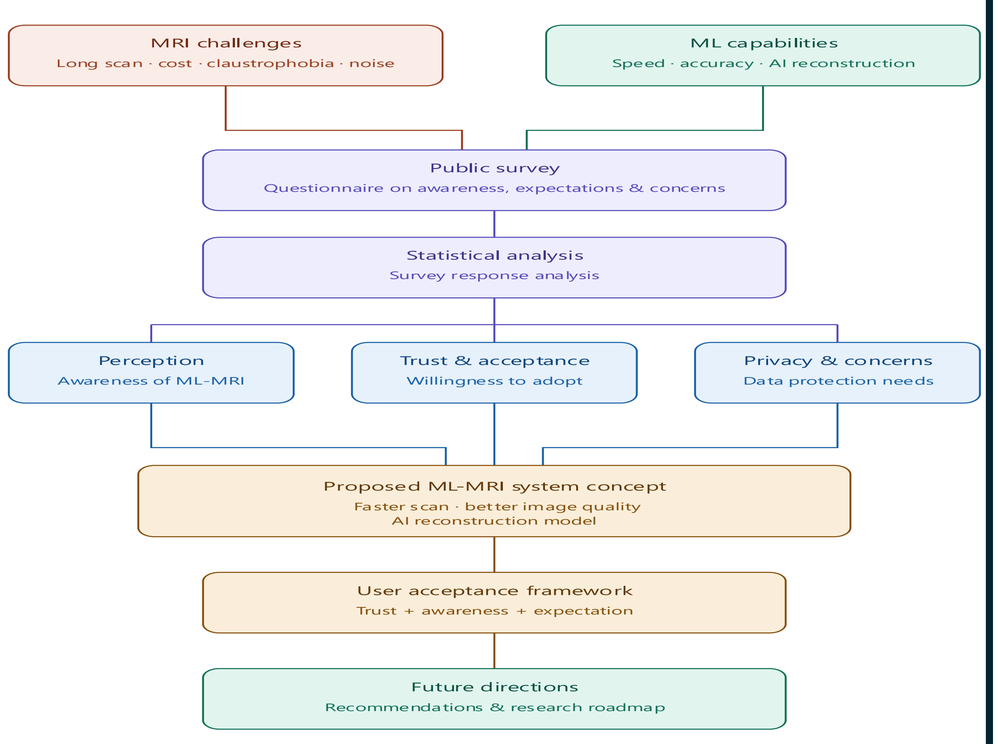
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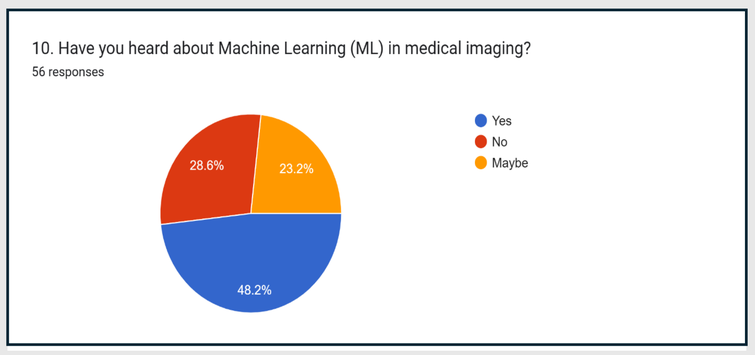
A significant association was found between trust in AI-MRI and data privacy concerns (χ^2 = 16.170, df = 8, p = 0.040). Respondents who trusted AI-enhanced MRI showed higher levels of concern regarding patient data usage.

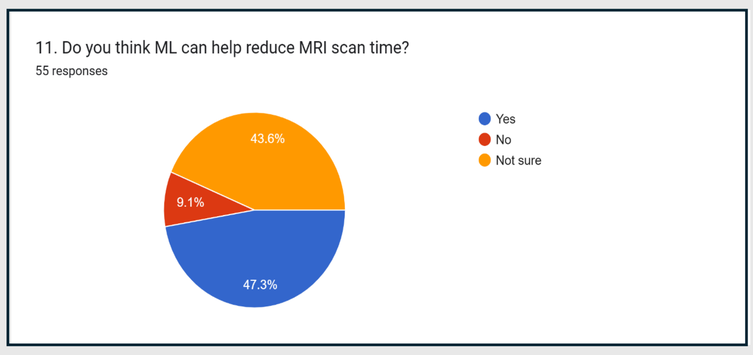


Expected data privacy assurance before using ML-driven MRI Strong encryption was the most preferred privacy assurance (37.5%), while a notable portion of respondents were unsure (32.1%). This suggests that although respondents demand technical data protection, a significant lack of awareness exists regarding privacy standards in ML-driven MRI systems.

WORK-FLOW







Result:

A significant positive correlation was found between prior MRI experience and the belief that MRI should be faster (r = 0.378, p = 0.004). Respondents who had undergone MRI scans were more likely to support faster MRI technology.

Results

Among 56 respondents, 55.4% expressed trust in AI-enhanced MRI, with faster scans (46.4%) and more accurate diagnosis (32.1%) as the most expected improvements. A significant association was found between AI-MRI trust and data privacy concern (χ^2 = 16.170, p = 0.040). Positive correlations were identified between prior MRI experience and desire for faster scans (r = 0.378, p = 0.004), and between ML awareness and confidence in scan time reduction (r = 0.327, p = 0.015).

Discussion

Although 48.2% had heard of ML in medical imaging, 43.6% remained unsure whether ML can reduce scan time, indicating a knowledge gap. The significant correlation between ML awareness and perceived utility suggests that public education could improve acceptance. The association between AI trust and privacy concern highlights that adoption depends not only on technological performance but also on transparent data protection practices.

Conclusion

Public expectations are broadly aligned with ML-MRI capabilities, yet adoption remains constrained by limited awareness and privacy concerns. Addressing these barriers through education, radiologist oversight, and clear data governance frameworks will be essential for successful integration of ML into clinical MRI practice.

Reference(s):

[1] Amin et al., 2022 — Brain Tumor Detection Survey.

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[3] Deep learning for accelerated and robust MRI reconstruction.

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